

SHEET	DRAWING	PURPOSE
E-101	V0.3 SYSTEM INTERCONNECT	Airflow, enclosed plasma subsystem, instruments, DAQ and safety
E-102	V0.3 HARDWIRED SAFETY CHAIN	Dual-channel concept, manual reset and monitored contactor feedback
E-201	G2B STATOR VALIDATION ONE-LINE	Polyphase source, phase measurement, field mapping and force stand
E-301	G2D HTS / STATOR INTEGRATION	Independent HTS protection and traveling-wave source
E-401	V5 ENERGY-SLED SINGLE-LINE	HV functional order, BMS, IMD, branch isolation and LV boundary
E-501	ARC POWER / DATA / SAFETY	Reasoning, mediation and independent protective paths
E-601	HARNESS ZONES / INTERFACES	Cable classes, routing separation, connector IDs and hold points

P2 RELEASES

Functional connections and independent safety boundaries
Equipment and connector identifiers
Cable-service classes and separation rules
De-energized assembly and inspection sequence

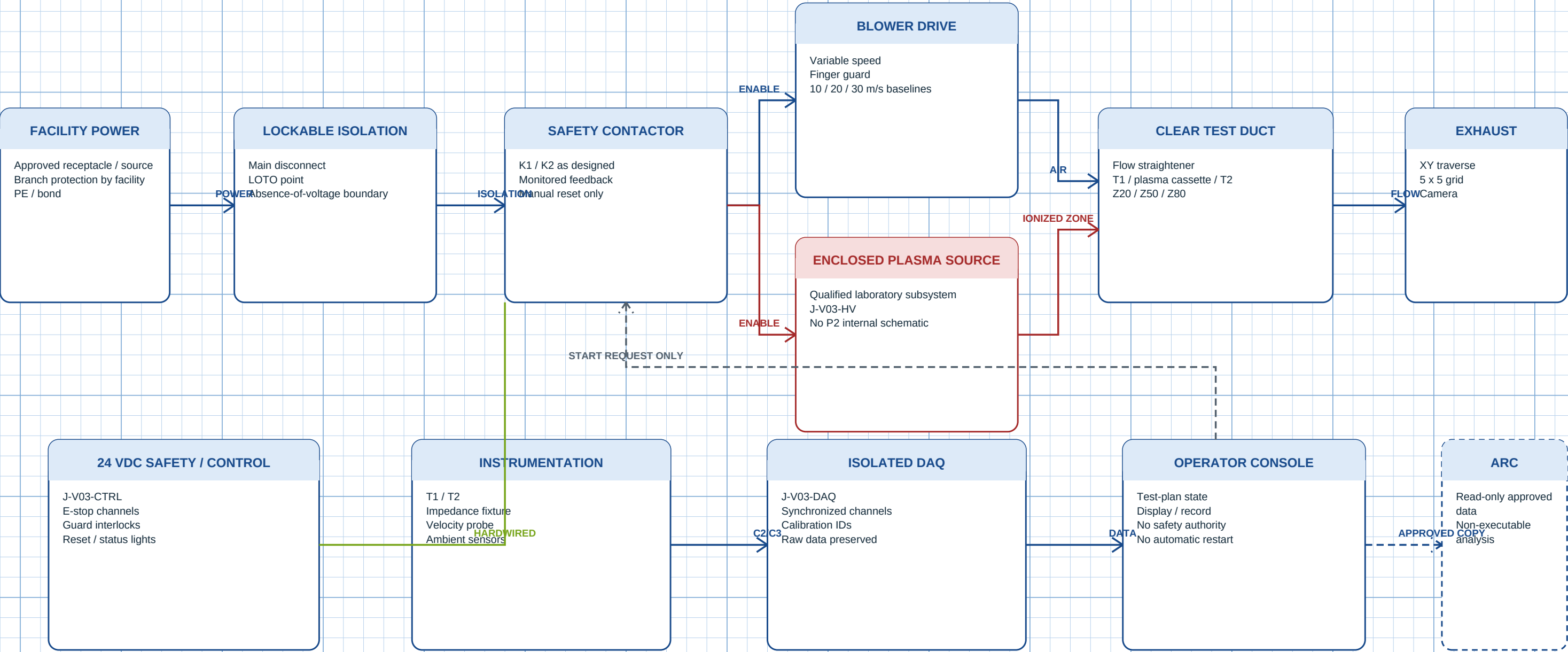
P2 DOES NOT RELEASE

Bus voltage or hazardous-energy limits
Power-wire gauge, fuse, contactor or coil rating
Plasma-pulser construction or energized procedure
Flight hardware, fabrication or certification approval

CORE RULE: ARC, DAQ AND ORDINARY SOFTWARE NEVER CARRY THE
HARDWIRED EMERGENCY-STOP OR PROTECTIVE-TRIP FUNCTION.

V0.3 MOVING-AIR TEST RIG - SYSTEM INTERCONNECT

100 x 100 mm duct | 10 / 20 / 30 m/s



NO PLASMA-ON TEST UNTIL THE ENCLOSED SOURCE, INTERLOCKS, LOTO, INSTRUMENTATION AND HOST-LAB PROCEDURE PASS THE READINESS REVIEW.

V0.3 MOVING-AIR TEST RIG - SYSTEM INTERCONNECT

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-101

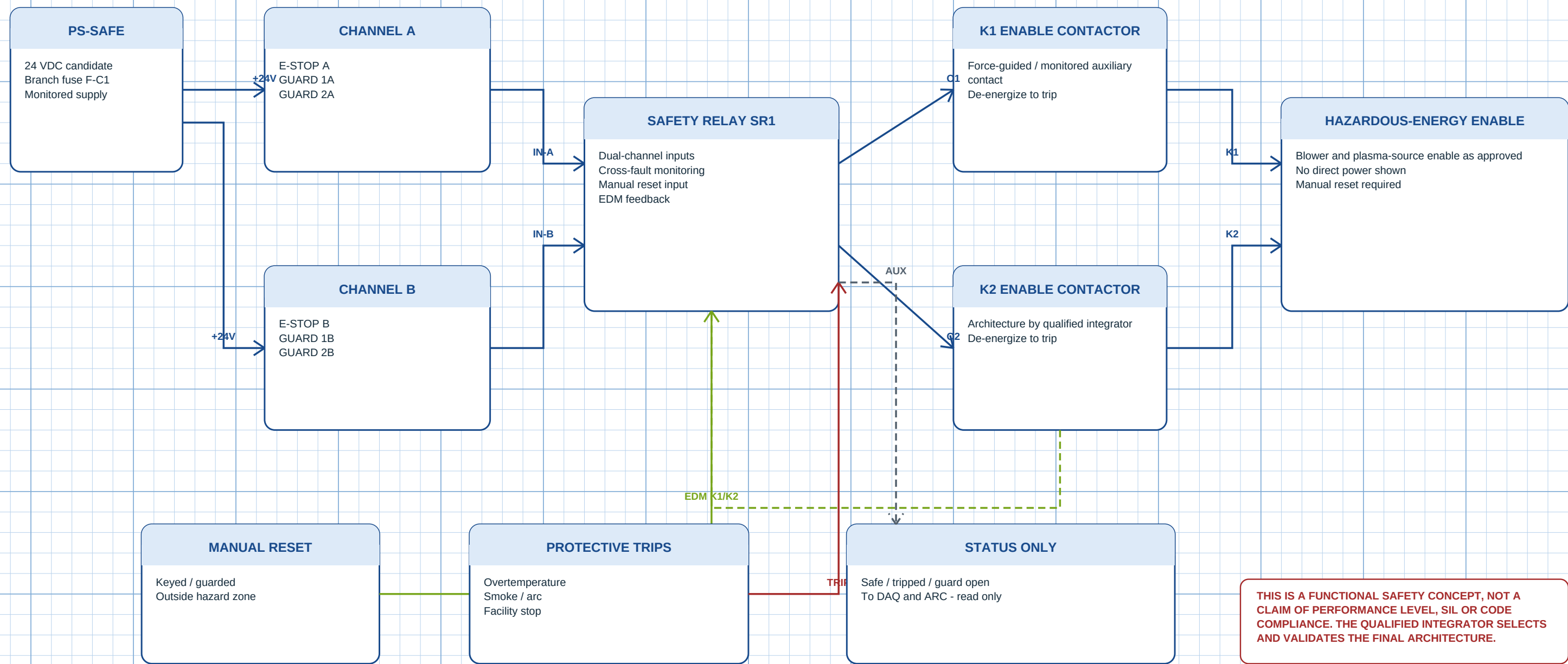
REV P2 | 2026-08-24

CRTFE V-2

CONCEPT ONLY

V0.3 HARDWIRED SAFETY CHAIN - CONCEPT

24 VDC low-energy logic shown; component category/rating TBD



V0.3 HARDWIRED SAFETY CHAIN - CONCEPT

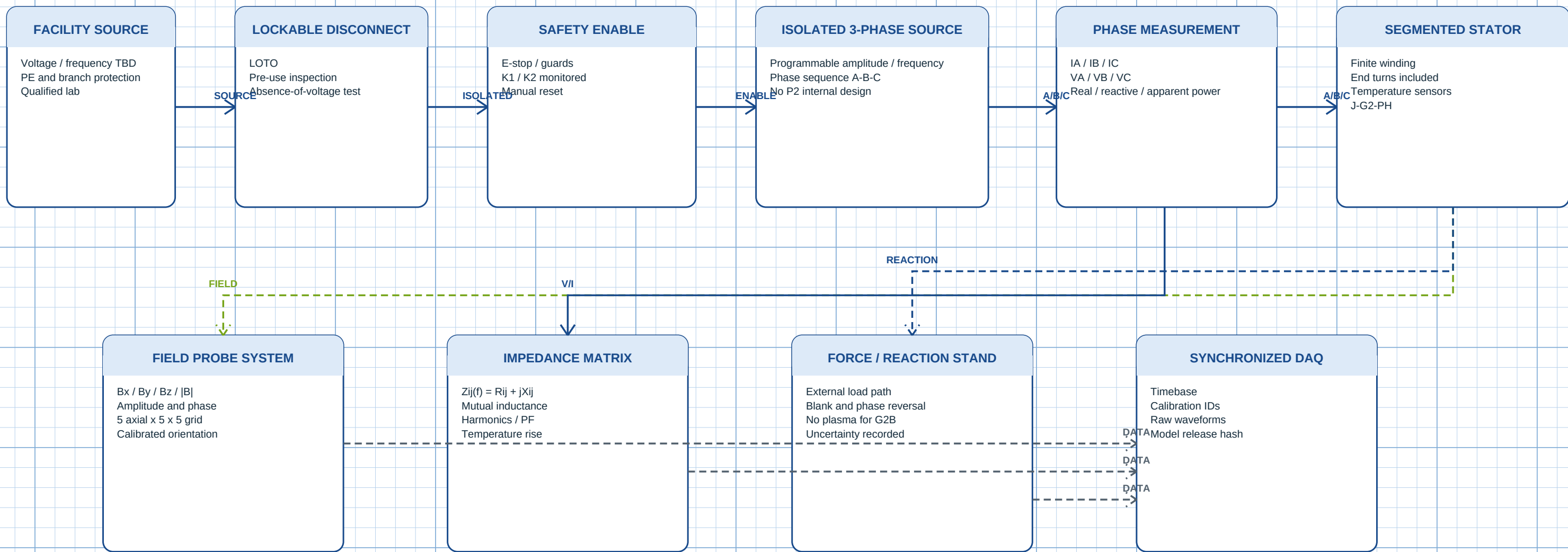
UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-102

REV P2 | 2026-08-24

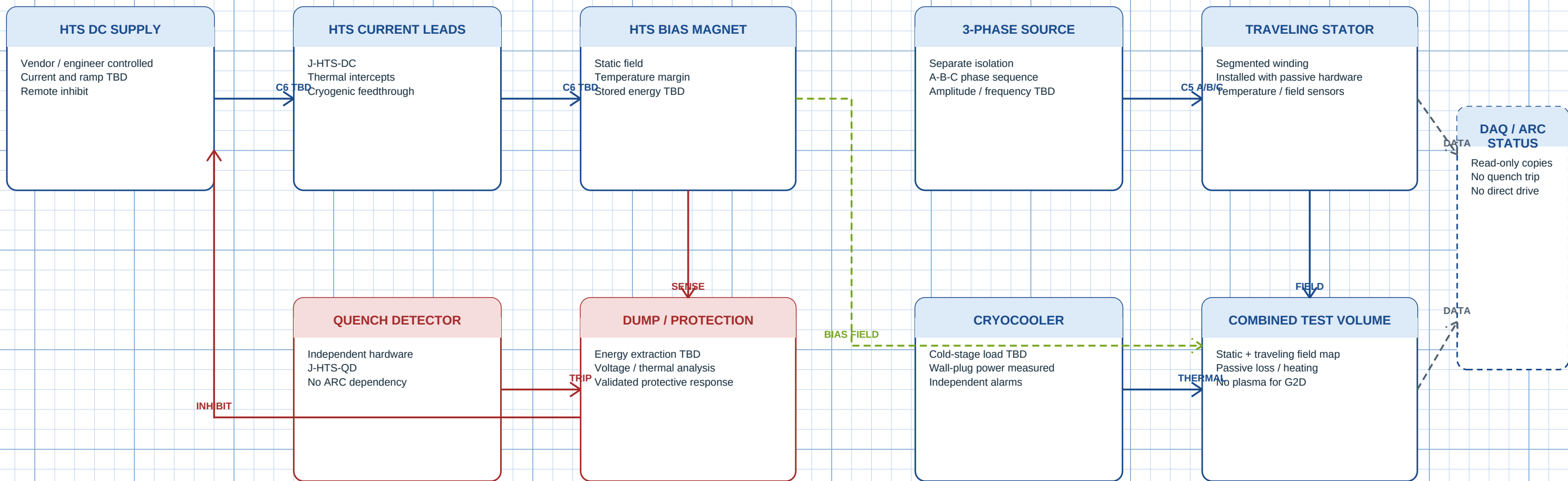
CRTFE V-2

CONCEPT ONLY



G2D COMBINED HTS BIAS AND TRAVELING-WAVE STATOR

plasma off | protection independent



STOP ON UNEXPLAINED HEATING, AC LOSS, FIELD DISTORTION, LOSS OF HTS TEMPERATURE MARGIN OR ANY QUENCH-PROTECTION ANOMALY.

G2D COMBINED HTS BIAS AND TRAVELING-WAVE STATOR

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-301

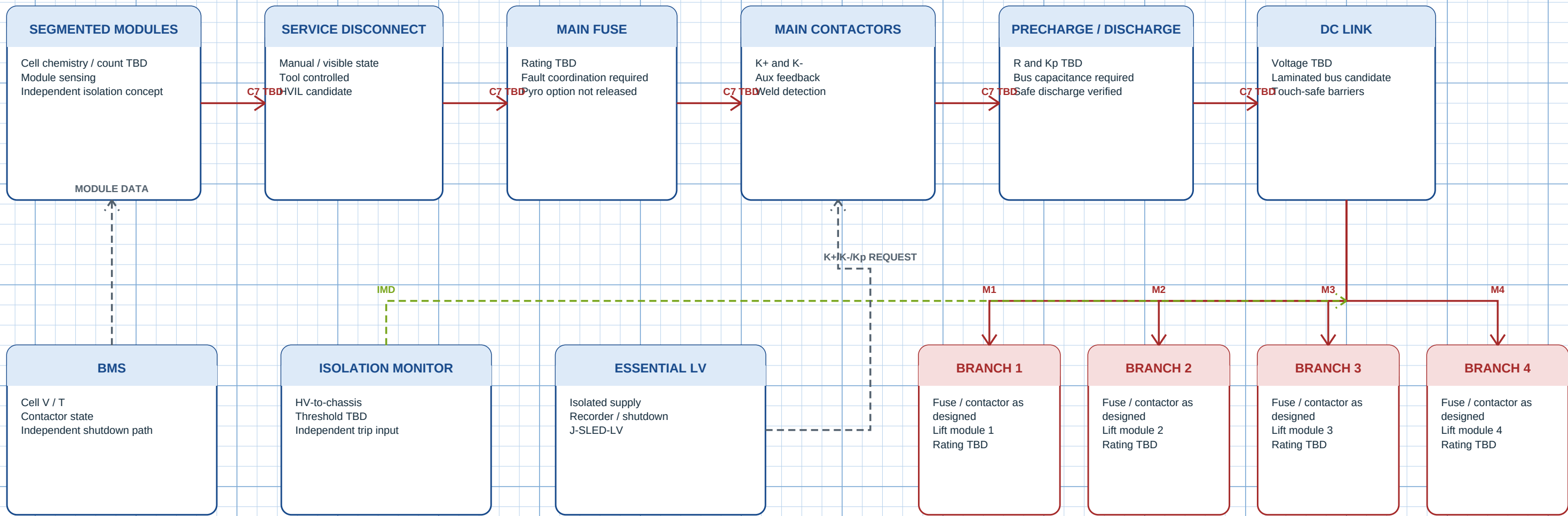
REV P2 | 2026-08-24

CRTFE V-2

CONCEPT ONLY

V5 TARGET ENERGY SLED - FUNCTIONAL SINGLE-LINE

1450 x 1050 x 340 mm envelope | 120 kg allowance



V5 TARGET ENERGY SLED - FUNCTIONAL SINGLE-LINE

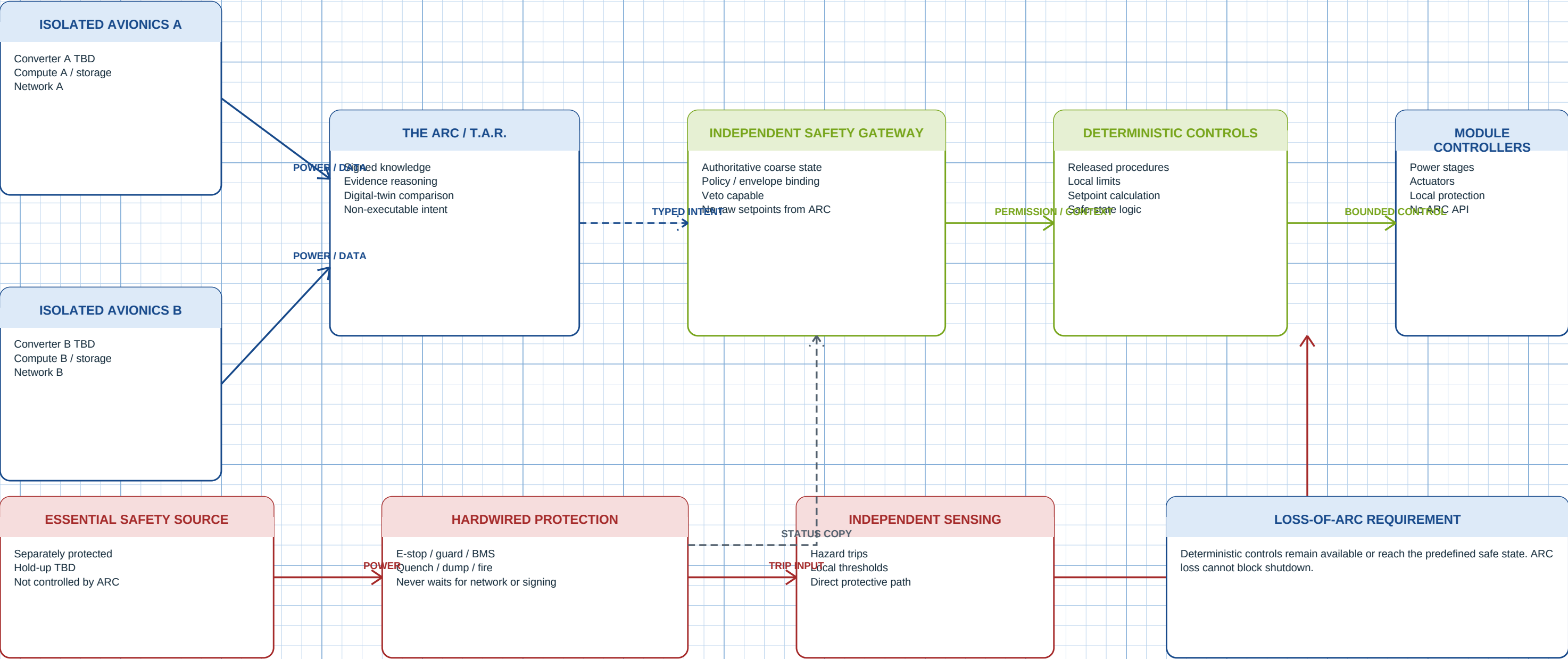
UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-401

REV P2 | 2026-08-24

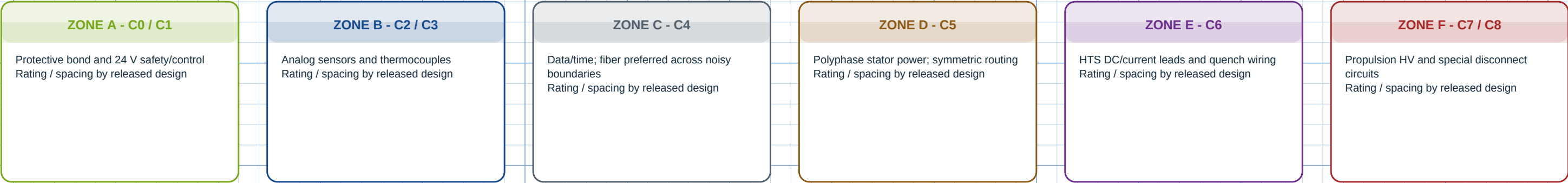
CRTFE V-2

CONCEPT ONLY



HARNESS ZONES, CABLE CLASSES AND RELEASE EVIDENCE

separate routing | controlled crossings | no inferred ratings



CONTROLLED CROSSING RULE

Cross power and sensor/data harnesses only at documented locations, preferably near 90 degrees. Maintain the released separation, shielding, bonding and enclosure-feedthrough design. Do not bundle unlike classes for convenience.

INTERFACE	SERVICE	MANDATORY FEATURES	RELEASE EVIDENCE
J-V03-CTRL	V0.3 low-voltage control	Keyed; status/control only	Netlist + low-energy test
J-V03-DAQ	Sensors / DAQ	Shielded pairs; no hazardous energy	Calibration + channel map
J-V03-HV	Enclosed plasma source	Lab-owned; touch-safe; interlocked	Facility approval
J-G2-PH	Stator A/B/C	Phase ID; symmetric routing	Impedance / thermal analysis
J-HTS-DC/QD	HTS current / quench	Independent protection	Magnet safety review
J-SLED-HV/LV	Energy sled	Keyed; HVIL candidate; LV segregated	Fault + insulation + FMEA
J-ARC-A/B	ARC redundant data/power	Isolated; no actuator pins	Interface control review

WIRE SIZE, OVERCURRENT PROTECTION, CONTACTOR RATINGS, CREEPAGE/CLEARANCE AND CONNECTOR PINOUTS REMAIN TBD UNTIL HARDWARE AND ELECTRICAL LOADS ARE SELECTED.

HARNESS ZONES, CABLE CLASSES AND RELEASE EVIDENCE

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-601

REV P2 | 2026-08-24

CRTFE V-2

CONCEPT ONLY